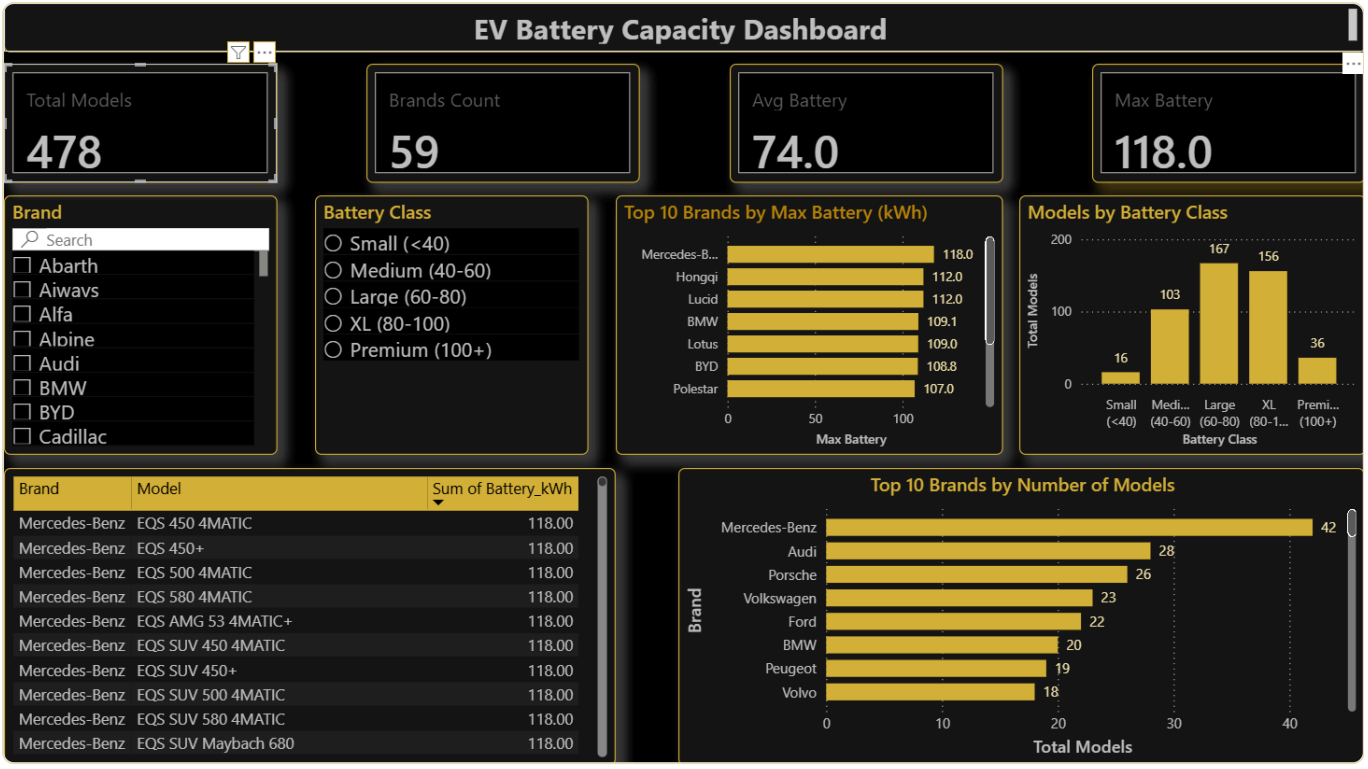
EV Battery Capacity Dashboard

Project documentation in questions and answers · Power BI

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The final dashboard in Power BI Desktop

Part 1 · About the project

1 What is this project?

An interactive **Power BI** dashboard that analyses the battery capacity (in kWh) of **478 electric-vehicle models from 59 brands**. A viewer can see at a glance which brands offer the biggest batteries, which brands have the widest range of models, and how all models are spread across battery-size classes.

2 What data was used?

The source is the Excel file `Battery_Capacity.xlsx`, sheet **Battery Report**. The main data sits in an Excel Table called `Battery_Report` (range A1:C479) with three columns: **Brand**, **model** and **Max Battery (kWh)**.

- 478 rows (one per model) and 59 brands.
- Capacity ranges from 21.3 to 118 kWh, with an average of about 74 kWh.
- The sheet also has a small side table (columns E:F) with the maximum per brand and a *Grand Total* row. It was **not** imported, so nothing is counted twice.

3 Which tools were used?

- **Excel**: source data.
- **Power BI Desktop**: report and visuals.
- **Power Query (M)**: cleaning and the Battery Class column.
- **DAX**: measures and the sort-order column.
- **JSON theme**: a custom black, gold and silver look.

4 What questions does the dashboard answer?

- How many models and brands are in the data, and what is the average and maximum battery size?
- Which 10 brands offer the highest maximum battery capacity?
- Which 10 brands have the most models?
- How many models fall in each battery class (Small to Premium)?
- What does the model list look like for one brand or one class?

5 How was the data prepared in Power Query?

1. Get Data, Excel, then choose the **Battery_Report** table (not the whole sheet).
2. Rename `Max Battery (kWh)` to `Battery_kWh` and `model` to `Model`; make sure `Battery_kWh` is a Decimal Number.
3. Fix the brand typed as `firefly` using **Replace Values** (to `Firefly`).
4. Replace the one empty model name (Firefly) with `Unknown` using a Custom Column.
5. Add a custom column `Battery Class` (M code below).
6. Close & Apply.

```
if [Battery_kWh] < 40 then "Small (<40)"
else if [Battery_kWh] < 60 then "Medium (40-60)"
else if [Battery_kWh] < 80 then "Large (60-80)"
else if [Battery_kWh] < 100 then "XL (80-100)"
else "Premium (100+)"
```

6 Which DAX measures were created, and why?

Five measures feed the KPI cards and the charts, and one calculated column keeps the classes in a logical order.

```
Total Models = COUNTROWS(Battery_Report)
Brands Count = DISTINCTCOUNT(Battery_Report[Brand])
Avg Battery  = AVERAGE(Battery_Report[Battery_kWh])
Max Battery  = MAX(Battery_Report[Battery_kWh])
Min Battery  = MIN(Battery_Report[Battery_kWh])
```

Class Order (calculated column). Without it, classes sort alphabetically (Large, Medium, Premium, Small, XL). We give each class a number and use *Column tools, Sort by column*.

```
Class Order = SWITCH(TRUE(),
    Battery_Report[Battery_kWh] < 40, 1,
    Battery_Report[Battery_kWh] < 60, 2,
    Battery_Report[Battery_kWh] < 80, 3,
    Battery_Report[Battery_kWh] < 100, 4,
    5)
```

7 Which visuals were used, and why?

Visual	Fields	Purpose
4 Cards	Total Models, Brands Count, Avg Battery, Max Battery	Headline numbers
Clustered bar	Brand, Max Battery (Top 10)	Brands with the biggest batteries
Clustered column	Battery Class, Total Models	Distribution of models by class
Clustered bar	Brand, Total Models (Top 10)	Brands with the widest range
Table	Brand, Model, Battery_kWh	Drill into individual models
2 Slicers	Brand, Battery Class	Filter the whole page

8 How does the interactivity work?

The two slicers filter every visual on the page. The Brand slicer has a search box (59 brands is too many to scroll) and multi-select; the Battery Class slicer shows the five classes in order. Clicking a bar also cross-filters the other visuals. The eraser icon on a slicer clears its filter.

9 How was the design chosen?

Black, gold and silver give a premium look that suits the topic. Instead of colouring each visual by hand, we wrote one **custom theme file** (`Black_Gold_Theme.json`) and imported it via *View, Themes, Browse for themes*. It sets the black page, dark cards with gold borders, gold bars and titles, and silver axis text. A few items (slicer text, card values) were then adjusted manually.

10 What are the key insights?

- The data has **478 models** from **59 brands**; the average battery is **74.0 kWh** and the maximum is **118 kWh**.
- **Mercedes-Benz** has both the most models (**42**) and the highest capacity (118 kWh). Audi (28) and Porsche (26) follow on model count.
- Next on maximum capacity: Hongqi and Lucid (112), BMW (109.1), Lotus (109), BYD (108.8).
- Class split: Large 167, XL 156, Medium 103, Premium 36, Small 16. Most models sit between 60 and 100 kWh.
- Only **36 models** exceed 100 kWh, so Premium is a niche segment.

11 How do I open and use the project from GitHub?

Suggested repository layout:

```
EV-Battery-Dashboard/
├── EV_Battery_Dashboard.pbix      (Power BI report)
├── data/Battery_Capacity.xlsx     (source data)
├── theme/Black_Gold_Theme.json    (custom theme)
├── images/dashboard.png           (screenshot)
└── docs/                          (these PDF guides)
```

1. Download the repository and open the `.pbix` file in Power BI Desktop.
2. If Power BI cannot find the data, go to *Home, Transform data, Data source settings* and point the source to your local copy of `Battery_Capacity.xlsx`.
3. To reuse the look on another report: *View, Themes, Browse for themes*, then pick the JSON file.

Part 2 · Errors we hit while building it, and how we fixed them

1 Duplicate summary data in the Excel sheet

Why it happened: The sheet has a second table (E:F) with per-brand maximums and a Grand Total row, so importing the whole sheet would double count.

How we fixed it: In the Navigator we selected the Excel Table `Battery_Report` (A1:C479) instead of the sheet, then pressed Transform Data.

2 "Capitalize Each Word" was missing, and it was the wrong tool anyway

Why it happened: We wanted to fix the lowercase brand `firefly`, but that command would also turn `BMW` into `Bmw` and `BYD` into `Byd`.

How we fixed it: Used **Replace Values** on the Brand column: `firefly` to `Firefly`. Only that one value changes.

3 Conditional Column was not used; Custom Column opened instead

Why it happened: The Custom Column dialog was opened by mistake, and it showed "The formula is incomplete".

How we fixed it: We typed the classification as an `if / else if` expression in M. Syntax is case-sensitive (lowercase `if`, `then`, `else`).

4 Two tables appeared in the Data pane

Why it happened: Besides the cleaned `Battery_Report`, an extra table with the original columns (`Max Battery (kWh)`, `model`) had also been loaded.

How we fixed it: In Power Query we deleted the extra query, then Close & Apply. Only one clean table remained.

5 DAX error: "The syntax for 'Models' is incorrect"

Why it happened: Power BI pre-fills `Measure =` in the formula bar. We pasted the whole formula after it, giving `Measure = Total Models = ...`.

How we fixed it: Delete the pre-filled text and type only `Name = formula`, for example `Total Models = COUNTROWS(Battery_Report)`.

6 Inconsistent decimals on cards (118.00 next to 74.0)

Why it happened: Max Battery used the default format.

How we fixed it: Measure tools: Format = Decimal number, Decimal places = 1.

7 Wrong chart type: 100% stacked bar instead of clustered bar

Why it happened: We pointed at the wrong icon. The axis showed 0% to 100% and only three brands appeared.

How we fixed it: Icons in the first row are: Stacked bar, Stacked column, **Clustered bar** (third), Clustered column, 100% stacked bar, 100% stacked column. Hover to confirm the name.

8 "This field can't be used here because a non-measure field is required"

Why it happened: In a column chart the X-axis is the category; we tried to drop the measure `Total Models` there. A stray *Count of Brand* was also on the axis.

How we fixed it: Rule: in a **bar** chart the category (Brand) goes on Y and the measure on X; in a **column** chart it is the opposite. We removed the stray field and rebuilt the visual.

9 Top N filter could not be found

Why it happened: The filter lives in the Filters pane, which was collapsed on the right.

How we fixed it: Select the visual, open Filters, expand the Brand card, Filter type = Top N, Top 10, By value = the measure, then **Apply filter**.

10 Duplicate and unreadable titles

Why it happened: The long title had been typed into the Y-axis title as well as the visual title. After applying the dark theme, one title became dark grey on black.

How we fixed it: Turned off the axis title, set the visual title once, and changed its font colour to gold.

11 Battery classes sorted alphabetically

Why it happened: Text columns sort A to Z by default, giving Large, Medium, Premium, Small, XL.

How we fixed it: Created the `Class Order` column with SWITCH, used *Sort by column*, then *Sort axis, ascending* on the visual.

12 Slicer options differed from what we expected

Why it happened: A Button slicer was used at first, which has no Style option; the new list slicer has no Dropdown style or Horizontal option in this version. It also started in Single select mode.

How we fixed it: Switched to the normal list slicer with its Search box, turned Single select off for multi-select, and used the eraser icon to clear. Horizontal layout was skipped as non-essential.

13 Cards suddenly showed 5 models and 1 brand

Why it happened: BMW and Premium were still selected in the slicers, so the whole page was filtered.

How we fixed it: Click the eraser on each slicer. Everything returns to 478 / 59 / 74.0 / 118.0.

14 Meaningless Total row in the table

Why it happened: The table summed all battery capacities (35,392.90 kWh), which has no meaning. It was also sorted by Model.

How we fixed it: Format visual, Totals = Off, and sort by *Battery_kWh* descending.

15 Card text colours would not follow the theme

Why it happened: Card labels and values have their own colour settings.

How we fixed it: Format visual, Callout value (number) and Category label (name): gold for numbers, silver for labels. The Format painter copies the style to the other cards.

16 One brand (Firefly) had an empty model name

Why it happened: The source has no model for Firefly. Replace Values with `null` did not work, because there the word is treated as text, not as an empty cell.

How we fixed it: Added a Custom Column `if [Model] = null or Text.Trim([Model]) = "" then "Unknown" else [Model]`, removed the old Model column and renamed the new one to `Model`. The table now shows **Unknown**.

Part 3 · Limitations and next steps

Known limitations

- The dataset only has maximum battery capacity. It has no price, range or year, so deeper comparisons are not possible yet.
- The Top 10 charts show brand-level maximums; they do not account for how many models a brand has.

Next steps

- Add range (km), price and year columns, then compare capacity against range.
- Add a scatter chart: number of models (X) against average capacity (Y) per brand.
- Add a second page for brand deep-dives and a dynamic title using DAX.
- Publish to Power BI Service and share the link on LinkedIn and Kaggle.